Group Project Contract

Pharma Payments & Prescription : Joining CMS Open Payments with Medicare Part D

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DSAI 694 : University of San Francisco

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Project Summary

We will join the CMS Open Payments General Payments file (payments and transfers of value from drug and device manufacturers to clinicians) with the Medicare Part D Prescribers by Provider and Drug file on the National Provider Identifier (NPI). The goal is to measure how industry payments relate to prescribing volume, cost, and brand-name drug use. Both files cover program year 2024. All processing will be done in Apache Spark (PySpark 4.0) because together the raw files are about 12.5 GB and 43.5 million lines.

Project repository: `https://github.com/Morioh/pharma-payments-prescription.git`

1 . Data Inventory

Output from the `data/` folder:

```
$ du -sh *
4.1G    Medicare_Part_D_Prescribers_by_Provider_and_Drug_2024.csv
8.4G    OP_DTL_GNRL_PGYR2024_P06302026_06032026.csv

$ wc -l *.csv
 28023893 Medicare_Part_D_Prescribers_by_Provider_and_Drug_2024.csv
 15498688 OP_DTL_GNRL_PGYR2024_P06302026_06032026.csv
 43522581 total
```

N/B: Each line count includes one header line. The Open Payments file has free-text fields (e.g. `Contextual_Information`) that can contain line breaks inside quotes, so its true record count may be a little lower than its line count. We will confirm the count with `df.count()` after reading the file in Spark with `multiLine=True`. The data folder is excluded from version control through `.gitignore`.

Dataset	Size	Data rows	Columns
Medicare Part D Prescribers by Provider & Drug (2024)	4.1 GB	28,023,892	22
CMS Open Payments : General Payments (PY2024)	8.4 GB	≈15,498,687	91

2 . Schema and Sample Records

Spark inferred the data types (`inferSchema=True`) from the first 200,000 rows of each file. **Join key:** `Prscrbr_NPI` (Part D) = `Covered_Recipient_NPI` (Open Payments). NPIs are 10-digit IDs. Spark infers them as `int`, but some NPIs are larger than the 32-bit limit, so in the pipeline we will read both key columns as `string` (or `long`). In Part D, empty values and suppression flags (`#`, `*`) appear in the `GE65_*` columns, and we will treat them as nulls.

2.1 Medicare Part D Prescribers by Provider and Drug (22 columns)

Column	Spark type
Prscrbr_NPI	int
Prscrbr_Last_Org_Name	string
Prscrbr_First_Name	string
Prscrbr_City	string
Prscrbr_State_Abrvtn	string
Prscrbr_State_FIPS	int
Prscrbr_Type	string
Prscrbr_Type_Src	string
Brnd_Name	string
Gnrc_Name	string
Tot_Clms	int
Tot_30day_Fills	double
Tot_Day_Suply	int
Tot_Drug_Cst	double
Tot_Benes	int
GE65_Sprsn_Flag	string
GE65_Tot_Clms	int
GE65_Tot_30day_Fills	double
GE65_Tot_Drug_Cst	double
GE65_Tot_Day_Suply	int
GE65_Bene_Sprsn_Flag	string
GE65_Tot_Benes	int

2.2 CMS Open Payments : General Payments (91 columns)

Column	Spark type
Change_Type	string
Covered_Recipient_Type	string
Teaching_Hospital_CCN	int
Teaching_Hospital_ID	int
Teaching_Hospital_Name	string
Covered_Recipient_Profile_ID	int
Covered_Recipient_NPI	int
Covered_Recipient_First_Name	string
Covered_Recipient_Middle_Name	string
Covered_Recipient_Last_Name	string
Covered_Recipient_Name_Suffix	string

Column	Spark type
Recipient_Primary_Business_Street_Address_Line1	string
Recipient_Primary_Business_Street_Address_Line2	string
Recipient_City	string
Recipient_State	string
Recipient_Zip_Code	string
Recipient_Country	string
Recipient_Province	string
Recipient_Postal_Code	string
Covered_Recipient_Primary_Type_1	string
Covered_Recipient_Primary_Type_2	string
Covered_Recipient_Primary_Type_3	string
Covered_Recipient_Primary_Type_4	string
Covered_Recipient_Primary_Type_5	string
Covered_Recipient_Primary_Type_6	string
Covered_Recipient_Specialty_1	string
Covered_Recipient_Specialty_2	string
Covered_Recipient_Specialty_3	string
Covered_Recipient_Specialty_4	string
Covered_Recipient_Specialty_5	string
Covered_Recipient_Specialty_6	string
Covered_Recipient_License_State_code1	string
Covered_Recipient_License_State_code2	string
Covered_Recipient_License_State_code3	string
Covered_Recipient_License_State_code4	string
Covered_Recipient_License_State_code5	string
Submitting_Applicable_Manufacturer_or_Applicable_GPO_Name	string
Applicable_Manufacturer_or_Applicable_GPO_Making_Payment_ID	bigint
Applicable_Manufacturer_or_Applicable_GPO_Making_Payment_Name	string
Applicable_Manufacturer_or_Applicable_GPO_Making_Payment_State	string
Applicable_Manufacturer_or_Applicable_GPO_Making_Payment_Country	string
Total_Amount_of_Payment_USDollars	double
Date_of_Payment	string
Number_of_Payments_Included_in_Total_Amount	int
Form_of_Payment_or_Transfer_of_Value	string
Nature_of_Payment_or_Transfer_of_Value	string
City_of_Travel	string
State_of_Travel	string
Country_of_Travel	string
Physician_Ownership_Indicator	string
Third_Party_Payment_Recipient_Indicator	string
Name_of_Third_Party_Entity_Receiving_Payment_or_Transfer_of_Value	string
Charity_Indicator	string
Third_Party_Equals_Covered_Recipient_Indicator	string
Contextual_Information	string
Delay_in_Publication_Indicator	string
Record_ID	int
Dispute_Status_for_Publication	string
Related_Product_Indicator	string

Column	Spark type
Covered_or_Noncovered_Indicator_1	string
Indicate_Drug_or_Biological_or_Device_or_Medical_Supply_1	string
Product_Category_or_Therapeutic_Area_1	string
Name_of_Drug_or_Biological_or_Device_or_Medical_Supply_1	string
Associated_Drug_or_Biological_NDC_1	string
Associated_Device_or_Medical_Supply_PDI_1	string
Covered_or_Noncovered_Indicator_2	string
Indicate_Drug_or_Biological_or_Device_or_Medical_Supply_2	string
Product_Category_or_Therapeutic_Area_2	string
Name_of_Drug_or_Biological_or_Device_or_Medical_Supply_2	string
Associated_Drug_or_Biological_NDC_2	string
Associated_Device_or_Medical_Supply_PDI_2	string
Covered_or_Noncovered_Indicator_3	string
Indicate_Drug_or_Biological_or_Device_or_Medical_Supply_3	string
Product_Category_or_Therapeutic_Area_3	string
Name_of_Drug_or_Biological_or_Device_or_Medical_Supply_3	string
Associated_Drug_or_Biological_NDC_3	string
Associated_Device_or_Medical_Supply_PDI_3	string
Covered_or_Noncovered_Indicator_4	string
Indicate_Drug_or_Biological_or_Device_or_Medical_Supply_4	string
Product_Category_or_Therapeutic_Area_4	string
Name_of_Drug_or_Biological_or_Device_or_Medical_Supply_4	string
Associated_Drug_or_Biological_NDC_4	string
Associated_Device_or_Medical_Supply_PDI_4	string
Covered_or_Noncovered_Indicator_5	string
Indicate_Drug_or_Biological_or_Device_or_Medical_Supply_5	string
Product_Category_or_Therapeutic_Area_5	string
Name_of_Drug_or_Biological_or_Device_or_Medical_Supply_5	string
Associated_Drug_or_Biological_NDC_5	string
Associated_Device_or_Medical_Supply_PDI_5	string
Program_Year	int
Payment_Publication_Date	string

2.3 First 5 Rows : Medicare Part D

The rows are transposed such that each column of the table is one record.

Column	Row 1	Row 2	Row 3	Row 4	Row 5
Prscrbr_NPI	1003000126	1003000126	1003000142	1003000142	1003000142
Prscrbr_Last_Org_Name	Enkeshafi	Enkeshafi	Khalil	Khalil	Khalil
Prscrbr_First_Name	Ardalan	Ardalan	Rashid	Rashid	Rashid
Prscrbr_City	Bethesda	Bethesda	Toledo	Toledo	Toledo
Prscrbr_State_Abrvtn	MD	MD	OH	OH	OH
Prscrbr_State_FIPS	24	24	39	39	39
Prscrbr_Type	Internal Medicine	Internal Medicine	Anesthesiology	Anesthesiology	Anesthesiology
Prscrbr_Type_Src	Claim-Specialty	Claim-Specialty	Claim-Specialty	Claim-Specialty	Claim-Specialty
Brnd_Name	Abrysvo	Boostrix Tdap	Acetaminophen-Codeine	Amitriptyline Hcl	Baclofen
Gnrc_Name	Rsv Vacc, Pref A And Pref B/Pf	Diphth, pertuss(Acell), tet Vac	Acetaminophen With Codeine	Amitriptyline Hcl	Baclofen
Tot_Clms	20	12	24	47	47
Tot_30day_Fills	20	12	24	47	47
Tot_Day_Suply	20	12	628	1410	1395
Tot_Drug_Cst	5994.01	648.26	362.48	271.17	333.35
Tot_Benes	20	12	11	16	26
GE65_Sprsn_Flag	–	–	#	–	–
GE65_Tot_Clms	20	12	–	30	33
GE65_Tot_30day_Fills	20	12	–	30	33
GE65_Tot_Drug_Cst	5994.01	648.26	–	150.18	222.14
GE65_Tot_Day_Suply	20	12	–	900	975
GE65_Bene_Sprsn_Flag	–	–	*	*	#
GE65_Tot_Benes	20	12	–	–	–

2.4 First 5 Rows : Open Payments

The rows are transposed. 55 of the 91 columns are empty in all five records and are left out (e.g. teaching-hospital fields, middle names, product slots 2–5). A dash (–) means the value is empty.

Column	Row 1	Row 2	Row 3	Row 4	Row 5
Change_Type	UNCHANGED	UNCHANGED	UNCHANGED	UNCHANGED	UNCHANGED
Covered_Recipient_Type	Covered Recipient Non-Physici...	Covered Recipient Physician	Covered Recipient Physician	Covered Recipient Non-Physici...	Covered Recipient Physician
Covered_Recipient_Profile_ID	10593686	232750	9569889	11109983	4432051
Covered_Recipient_NPI	1790251015	1134115629	1417453861	1073086989	1942795463
Covered_Recipient_First_Name	AMY	CHRISTOPHER	ANDREA	ANGELA	ANNA
Covered_Recipient_Last_Name	PONZIO	STALLTER	TAI	NEWMAN	MCFARLAND
Recipient_Primary_Business_Street_Address_Line1	2700 E 29TH ST STE 305	2972 INDIAN TRAIL RD	1135 S SUNSET AVE STE 401	2900 WATKINS RD	5303 S CEDAR ST STE 205
Recipient_City	BRYAN	AURORA	WEST COVINA	BATTLE CREEK	LANSING
Recipient_State	TX	IL	CA	MI	MI
Recipient_Zip_Code	77802	60502	91790	49015	48911
Recipient_Country	United States	United States	United States	United States	United States
Covered_Recipient_Primary_Type_1	Nurse Practitioner	Doctor of Osteopathy	Medical Doctor	Nurse Practitioner	Doctor of Osteopathy
Covered_Recipient_Specialty_1	Physician Assistants & Advanc...	Allopathic & Osteopathic Phys...	Allopathic & Osteopathic Phys...	Physician Assistants & Advanc...	Allopathic & Osteopathic Phys...
Covered_Recipient_License_State_code1	TX	IL	CA	MI	MI

Column	Row 1	Row 2	Row 3	Row 4	Row 5
Submitting_Applicable_Manufacturer_or_Applicable_GPO_Name	Phadia US Inc.	Phadia US Inc.	Phadia US Inc.	Phadia US Inc.	Phadia US Inc.
Applicable_Manufacturer_or_Applicable_GPO_Making_Payment_ID	100000010419	100000010419	100000010419	100000010419	100000010419
Applicable_Manufacturer_or_Applicable_GPO_Making_Payment_Name	Phadia US Inc.	Phadia US Inc.	Phadia US Inc.	Phadia US Inc.	Phadia US Inc.
Applicable_Manufacturer_or_Applicable_GPO_Making_Payment_State	MI	MI	MI	MI	MI
Applicable_Manufacturer_or_Applicable_GPO_Making_Payment_Country	United States	United States	United States	United States	United States
Total_Amount_of_Payment_USDollars	83.55	15.69	21.63	13.70	13.50
Date_of_Payment	10/23/2024	10/31/2024	05/02/2024	07/11/2024	02/12/2024
Number_of_Payments_Included_in_Total_Amount	1	1	1	1	1
Form_of_Payment_or_Transfer_of_Value	Cash or cash equivalent	Cash or cash equivalent	Cash or cash equivalent	Cash or cash equivalent	Cash or cash equivalent
Nature_of_Payment_or_Transfer_of_Value	Food and Beverage	Food and Beverage	Food and Beverage	Food and Beverage	Food and Beverage
Physician_Ownership_Indicator	No	No	No	No	No
Third_Party_Payment_Recipient_Indicator	No Third Party Payment	No Third Party Payment	No Third Party Payment	No Third Party Payment	No Third Party Payment
Delay_in_Publication_Indicator	No	No	No	No	No
Record_ID	1067575426	1067577204	1067575510	1067575608	1067575712
Dispute_Status_for_Publication	No	No	No	No	No
Related_Product_Indicator	Yes	Yes	Yes	Yes	Yes
Covered_or_Noncovered_Indicator_1	Covered	Covered	Covered	Covered	Covered
Indicate_Drug_or_Biological_or_Device_or_Medical_Supply_1	Device	Device	Device	Device	Device
Product_Category_or_Therapeutic_Area_1	Allergy/ Immunology	Allergy/ Immunology	Allergy/ Immunology	Allergy/ Immunology	Allergy/ Immunology
Name_of_Drug_or_Biological_or_Device_or_Medical_Supply_1	ImmunoCAP	ImmunoCAP	ImmunoCAP	ImmunoCAP	ImmunoCAP
Program_Year	2024	2024	2024	2024	2024
Payment_Publication_Date	06/30/2026	06/30/2026	06/30/2026	06/30/2026	06/30/2026

3 . Research Questions and Planned Spark Operations

Each question needs both datasets: the payment side comes only from Open Payments, and the prescribing side comes only from Part D.

Q1. Do clinicians who receive industry payments prescribe more brand-name drugs and spend more per claim than clinicians in the same specialty who receive no payments?
Spark: `filter` (keep rows with a valid NPI and physician/NPP recipient types); `map` to `(npi, amount)` and `reduceByKey` to get total payments per NPI; `map`/`reduceByKey` on Part D to get claims, brand claims (`Brnd_Name` $\neq$ `Gnrc_Name`) and cost per NPI; `leftOuterJoin` from prescribers to payments, so that unpaid prescribers are kept; `groupBy(Prscrbr_Type, paid_flag)` with `agg(avg)`.

Q2. When a manufacturer pays a clinician in connection with a specific drug, does that clinician prescribe more of that drug than unpaid peers do?
Spark: `flatMap` to turn the five `Name_of_Drug_or_Biological_..._1-5` slots into separate `(npi, drug)` records; `map` to normalize drug names (upper case, trim); `join` on the composite key `(npi, drug)` against Part D `Brnd_Name`; `reduceByKey` to sum claims; `distinct` to count the paid and unpaid prescribers of each drug.

Q3. Is there a dose–response relationship? How do prescribing volume and cost change across payment-size buckets and payment types (Food & Beverage, Consulting, Travel, Speaker fees)?
Spark: `reduceByKey` on `((npi, Nature_of_Payment), amount)`; `map` to put each NPI's total into a bucket (\$0, <\$100, \$100–\$1k, \$1k–\$10k, >\$10k); `join` with per-NPI Part D totals; `aggregateByKey` to compute the mean and variance for each bucket; `sortByKey`.

Q4. Which states have the largest share of Part D prescribers receiving payments, and is that share associated with higher drug cost per beneficiary?
Spark: `join` on NPI; `map` to `(state, (paid, cost, benes))`; `reduceByKey` for each state's totals; `broadcast` join with a small state lookup table; `sortBy` and `corr` to compute the state-level correlation.

Q5. Which manufacturers pay the most per claim of their drugs prescribed in Part D, and which specialties do they target?
Spark: `reduceByKey` on `((manufacturer, drug), amount)`; `join` with Part D claims summed by drug; `map` to compute payment-per-claim ratios; `takeOrdered` / `top` to get the top-k manufacturers; `groupByKey` to get each top manufacturer's specialty breakdown.

N/B: We figured to convert both CSVs to Parquet to make later reads fast; `cache` the aggregated tables per NPI that several questions reuse; repartition both sides on NPI before large joins; check NPI match rates and null keys before running any analysis.

4 . Roles, Responsibilities, and Weekly Plan

Maurice owns the *Open Payments* pipeline: ingestion, cleaning, and payment aggregates. **Miguel** owns the *Medicare Part D* pipeline: ingestion, cleaning, prescriber/drug aggregates, and drug-name normalization. Both share responsibility for the join, validation, and final deliverables. All code goes through GitHub pull requests, and the other member reviews each one before it is merged.

Week	Member	Task	Deliverable
Wk 1 Nov 1–7	Maurice	Create the GitHub repo, `.gitignore` and folder structure; download the Open Payments PY2024 file; set up the shared PySpark environment.	Repo on GitHub; Open Payments CSV in `data/`
	Miguel	Download the Part D 2024 file; read both CMS data dictionaries; profile file sizes, row counts and schemas.	Part D CSV in `data/`; schema notes
	Both	Kickoff meeting; choose the research questions; draft this contract.	Contract draft
Wk 2 Nov 8–14	Maurice	Ingest Open Payments with an explicit schema (NPI as string, parsed dates); filter to clinician recipients; write to Parquet. Ingest the full Part D file in Spark with an explicit schema; convert suppression flags to nulls; write to Parquet; document column definitions from the CMS data dictionary.	`ingest_open_payments.py`; `ingest_partd.py`; Parquet output for both files; row-count check; data dictionary notes
	Miguel	Profile both files from CMS API samples (Part D: 22 columns; Open Payments: 500 rows, 91 columns); test NPI as the join key on 20 sample NPIs (15 matched, 233 Part D drug rows); draft the schema and join plan.	`notebooks/01_medicare_inspection`, `02_payments`, `03_join_validation`; `docs/initial_sample_dataset_report.md`
	Both	Sign this contract; agree on coding conventions and the PR review process; review the sample join plan (aggregate each side to one row per NPI before joining, which avoids a many-to-many join, for Q1, Q3 and Q4).	Signed contract; agreed join plan
Wk 3 Nov 15–21	Maurice	Build per-NPI payment aggregates (total, count, by nature, by manufacturer); explode the five product slots into a long table.	Cached payment-aggregate tables
	Miguel	Build per-NPI prescribing aggregates (claims, cost, beneficiaries, brand share); normalize drug names for matching.	Prescriber- and drug-level aggregate tables
	Both	Validate the join: NPI match rate, duplicate and null keys, how well drug names match across the two files.	Data-quality / match-rate report (1–2 pages)
Wk 4 Nov 22–28 *(Thanksgiving)*	Maurice	Q1 (paid vs. unpaid prescribing by specialty) and Q3 (payment buckets and nature of payment).	Notebook with Q1 and Q3 results and draft figures

Week	Member	Task	Deliverable
	Miguel	Q2 (drug-specific payment → prescribing join on `(npi, drug)`).	Notebook with Q2 results and draft figures
Wk 5 Nov 29–Dec 5	Maurice	Q4 (state-level payment share vs. cost per beneficiary).	Q4 results and map/plot
	Miguel	Q5 (manufacturer payment-per-claim ratios and specialty targeting).	Q5 results and top-k tables
	Both	Tune performance (partitioning, broadcast joins, caching); review each other's code.	Merged, reproducible pipeline on `main`
Wk 6 Dec 6–12	Maurice	Write the report sections on data, methods, and Q1/Q3/Q4 findings; write limitations (correlation vs. causation, data suppression).	Report draft v1
	Miguel	Write the report sections on Q2/Q5 findings; produce the final visualizations; build the slide deck.	Final figures; slide deck draft
	Both	Edit the full report draft together.	Report draft v2
Wk 7 Dec 13–16	Maurice	Final proofreading; README with instructions for reproducing the pipeline.	Final report (PDF); README
	Miguel	Finalize the slides; clean up the repo and add run scripts.	Final slides; tagged release
	Both	Rehearse the presentation; **submit by Dec 16**.	Submitted project

5 . Online Meeting Schedule

We will huddle on Slack (or Google Meet otherwise) and use Slack's group messaging for day-to-day communication. All times are Pacific Time. Meeting notes and action items go in `docs/meetings.md` in the repo.

Meeting	When	Purpose
Weekly planning	Sundays, 7:00 – 8:00 am	Review last week's work, assign tasks for the coming week
Mid-week check-in	Wednesdays, 7:30 – 8:00 am	Report progress, unblock issues, review PRs

Date	Type	Focus
Sun Nov 6	Planning	Kickoff; choose questions; split data download
Wed Nov 7	Check-in	Data downloaded; schema review; contract draft
Sun Nov 8	Planning	Sign contract; start ingestion
Wed Nov 11	Check-in	Ingestion progress
Sun Nov 15	Planning	Parquet outputs done; start aggregates
Wed Nov 18	Check-in	Join validation
Sun Nov 22	Planning	Match-rate report; assign Q1–Q3
Wed Nov 25	Async	Written update in group chat (Thanksgiving)
Sun Nov 29	Planning	Q1–Q3 results review; assign Q4–Q5
Wed Dec 2	Check-in	Performance tuning; code review
Sun Dec 6	Planning	Results freeze; report outline
Wed Dec 9	Check-in	Report draft review
Sun Dec 13	Planning	Final edits; slide review
Tue Dec 15	Rehearsal	Full presentation run-through
Wed Dec 16	—	**Final submission**

Either member can reschedule a meeting with at least 24 hours' notice. If a deliverable is going to be late, the owner tells the other member by the next check-in and proposes a new date.

6 . Agreement

By signing below, each member agrees to the roles, responsibilities, timeline, and meeting schedule in this contract, and agrees to communicate promptly if anything changes.

Name	Signature	Date
Maurice		November 8, 2026
Miguel		November 8, 2026

NOTE: We used AI for the formatting of our tables on this contract.